

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. INTRODUCTION TO THE PLANNING PHASE:

The Planning Phase serves as the structural roadmap for the ComicCraft application. Developing a system that seamlessly integrates a local web framework (FastAPI) with highly complex external AI endpoints (Google Gemini and Hugging Face) requires meticulous scheduling. A structured, phased approach was adopted to ensure that networking, logic, and UI design were developed and tested systematically without bottlenecks.

3. AGILE DEVELOPMENT LIFECYCLE :

To maximize efficiency, the team abandoned a rigid waterfall model in favor of an agile, modular development lifecycle. The project was broken down into five distinct execution phases, allowing the team to iteratively build, test, and refine individual components before linking them together.

4. PHASED EXECUTION ROADMAP

Phase 1: Environment & Tooling Selection

- **Objective:** To finalize the technical stack and prepare the local workspace.
- **Execution:** Conducted comparative analysis to select Gemini 1.5 Flash (for JSON structuring) and Stable Diffusion v1.5 (for visual rendering). Set up a strict Python virtual environment (venv) to isolate dependencies and established the MVC folder hierarchy (app, static, templates).

Phase 2: Core AI Scripting (Backend Models)

- **Objective:** To program the specific Python modules responsible for communicating with cloud AI services.

- **Execution:** Drafted the `gemin_i_handler.py` to securely transmit prompts and parse the returned JSON/text data. Developed the `image_synthesis.py` script to handle byte-stream downloads from Hugging Face. Initiated the mathematical layout logic for the FPDF compiler.

Phase 3: Web Server & Routing Integration

- **Objective:** To build the central nervous system of the application using FastAPI.
- **Execution:** Authored the controller script (`app_routes.py`) to manage HTTP POST requests from the user. Designed the asynchronous execution pipeline ensuring that the AI generation happens sequentially (Text → Images → Layout → PDF) without freezing the web server.

Phase 4: Frontend & Templating Setup

- **Objective:** To craft a frictionless, intuitive user interface.
- **Execution:** Designed the HTML5 landing page with necessary parameter dropdowns. Integrated the Jinja2 templating engine to dynamically parse backend python arrays and display the final comic strip seamlessly on the browser screen.

Phase 5: Quality Assurance & Deployment

- **Objective:** To validate system stability and prepare the final demonstration.
- **Execution:** Booted the Uvicorn ASGI server for local testing. Verified error handling mechanisms, ensured the FPDF compiler correctly formatted documents without image distortion, and finalized the project documentation.

5. RISK MANAGEMENT & BOTTLENECKS :

Proactive planning required anticipating technical failures, particularly regarding third-party APIs.

- **Risk 1: Cloud Server Timeouts.** External AI models can experience high traffic, leading to HTTP 503 errors.
 - *Mitigation:* The backend was coded with strict `try-except` blocks to catch these errors and reroute the image generation task.
- **Risk 2: Local Network Disruptions.** Total internet loss would crash the PDF compiler if images failed to download.
 - *Mitigation:* Engineered a localized Python PIL script that instantly draws a backup placeholder image with text, ensuring the application always yields a finished PDF regardless of network status.

6. RESOURCE ALLOCATION:

Work was systematically distributed among team members. Tasks were divided into backend API routing, Jinja2 frontend templating, PDF coordinate mathematics, and rigorous network failure testing, ensuring a balanced and rapid development cycle.